STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 14: Waves

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Perpendicular.

Longitudinal.

v=flambda.

2pi/lambda.

Resultant displacement is the algebraic sum.

Pulse is inverted.

|f_1-f_2|.

Maximum displacement from equilibrium.

100 m/s.

2 m.

4 Hz.

Doubles.

No.

+x direction.

30Hz; 1/30s.

+2 cm.

4 Hz; .25 s.

Particles oscillate locally while the disturbance propagates.

30m/s; lambda=6m.

Alternating loudness occurs at 2 beats per second.